

RATL Specification

RATL (R-based Array Manipulation Language) is a stack-based, esoteric programming language inspired by [MATL](#). It leverages R's powerful statistical and matrix capabilities through a concise, postfix syntax, making it highly suitable for code golf.

Total symbols: 419 (all 2-byte or shorter)

1. Introduction

RATL is implemented in R and uses R's internal data structures (vectors, lists, matrices) to handle stack operations. It translates RATL code into R code, which is then evaluated.

1.1 The Name

The name stands for **R-based Array Manipulation Language**, reflecting its foundation in R and its primary focus on array-based operations.

1.2 Notation

- **\$n**: Represents the n -th element popped from the stack (starting from the top).
 - **stack**: The global list used for storage.
 - **H, G, L, M**: Clipboards for temporary storage outside the stack.
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2. The Stack and Data Types

RATL is a stack-based language where every operation interacts with a global stack (an R `list`).

Data Types

RATL uses native R types: - **Numeric**: Integers and floating-point numbers (R `numeric`). - **Logical**: Boolean values (`TRUE`, `FALSE`). - **Character**: Strings of text (R `character`). - **List/Cell Array**: Collections of items, which can be of mixed types (R `list`). - **Matrix**: 2D arrays (R `matrix`).

3. Statements and Separators

- **Space:** Acts as a token separator.
 - **Newline:** Ignored by the parser but acts as a token separator.
 - **Comments:** Lines starting with `#` are treated as comments and ignored.
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4. Literals

Literals are pushed onto the stack as soon as they are encountered.

4.1 Numbers

Standard numeric notation: 1, 10, 3.14, -5.

4.2 Numerical Arrays (Vectors)

Enclosed in square brackets `[...]`. Elements are separated by spaces. Example: `[1 2 3]` pushes a numeric vector of length 3.

4.3 Character Arrays (Strings)

Enclosed in single quotes `'...'`. Example: `'hello'` pushes a character vector of length 1.

4.4 Cell Arrays (Lists)

Enclosed in curly braces `{...}`. Example: `{1 'a' [1 2]}` pushes an R list containing three different types.

5. Control Flow

RATL uses special symbols for conditional branching and loops.

5.1 Conditional Branching (? ...])

- **Symbols:** ? starts the block,] ends it.
- **If-Else:** ? `truthy` ; `falsy`]. The ; separator distinguishes the “then” and “else” branches.
- **Logic:** Pops the top element. If it’s true, it executes the code until] (or ;). If false and ; is present, it executes the code between ; and].

5.2 While/Repeat Loop (" ... ")

- **Symbol:** "
- **Logic:** Starts a loop. At the end of the loop (marked by another "), it pops the top element. If it’s true, it repeats.

5.3 For-Each Loop ((...))

- **Symbols:** (and)
- **Logic:** (pops an array/list. It then iterates through each element, pushing it onto the stack and executing the code between (and).

5.4 Infinite Loop (` ... `)

- **Symbols:** ` starts and ends the loop.
- **Logic:** Executes the code within the backticks repeatedly. Use the **X** (break) symbol to exit.

5.5 Higher-Order Functions (Block Consumers)

These functions pop a {block} from the stack and apply it to data.

Symbol	Name	Description
q	Map	{block} array q — applies block to each element
e	Filter	{block} array e — keeps elements where block returns truthy
y	Reduce	{block} array y — fold left with block
z	Repeat	N {block} z — executes block N times
@	Execute	{block} @ — pops and executes a block immediately

6. Implicit Actions

6.1 Initial Actions

- The stack is initialized as an empty `list()`.
- A connection to `stdin` is opened for input operations.

6.2 Final Actions

- **Implicit Print:** If the stack contains exactly one element, it is printed automatically. If it contains more than one, the entire stack is printed as a list.
 - **Invisible Output:** The final evaluation result is returned invisibly in R.
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7. Symbol Reference

7.1 Arithmetic & Comparison

Symbol	Description	Stack Effect
!	Transpose	$1 \rightarrow 1$
%	Modulo	$2 \rightarrow 1$
*	Multiply	$2 \rightarrow 1$
+	Add	$2 \rightarrow 1$
-	Subtract	$2 \rightarrow 1$
/	Divide	$2 \rightarrow 1$
<	Less	$2 \rightarrow 1$
<=	LessEqual	$2 \rightarrow 1$
=	Equal	$2 \rightarrow 1$
>	Greater	$2 \rightarrow 1$
>=	GreatEqual	$2 \rightarrow 1$
^	Power	$2 \rightarrow 1$
ab	Abs	$1 \rightarrow 1$
cl	Ceiling	$1 \rightarrow 1$
fl	Floor	$1 \rightarrow 1$
ro	Round	$1 \rightarrow 1$
sg	Sign	$1 \rightarrow 1$

Symbol	Description	Stack Effect
sq	Sqrt	$1 \rightarrow 1$
tr	Trunc	$1 \rightarrow 1$
~	Not	$1 \rightarrow 1$

7.2 Stack & Control

Symbol	Description	Stack Effect
D	Duplicate	$1 \rightarrow 2$
Ls	Stack Length	$0 \rightarrow 1$
U	Unpack	$1 \rightarrow 0$
i	Input	$0 \rightarrow 1$
w	Swap	$2 \rightarrow 2$
x	Delete	$1 \rightarrow 0$

7.3 Higher-Order Functions

Symbol	Description	Stack Effect
Fq	Map (evaluator)	$0 \rightarrow 1$
Fr	Reduce	$2 \rightarrow 1$
Ft	Filter (evaluator)	$0 \rightarrow 1$
Fx	Repeat (evaluator)	$0 \rightarrow 1$
e	Filter (evaluator)	$0 \rightarrow 1$
q	Map (evaluator)	$0 \rightarrow 1$
y	Reduce (evaluator)	$0 \rightarrow 1$
z	Repeat (evaluator)	$0 \rightarrow 1$

7.4 Array Operations

Symbol	Description	Stack Effect
:	Sequence	$2 \rightarrow 1$
O	Sort	$1 \rightarrow 1$
R	Reverse	$1 \rightarrow 1$
SH	Head	$1 \rightarrow 1$
SR	Rank	$1 \rightarrow 1$
ST	Tail	$1 \rightarrow 1$

Symbol	Description	Stack Effect
e1	Extract Element [[	$2 \rightarrow 1$
en	Extract Name \$	$2 \rightarrow 1$
es	Extract Subset [	$2 \rightarrow 1$
fE	FirstElem	$1 \rightarrow 1$
fu	Flatten	$1 \rightarrow 1$
hd	HeadN	$2 \rightarrow 1$
ix	IndexOf	$2 \rightarrow 1$
l	Length	$1 \rightarrow 1$
la	LastElem	$1 \rightarrow 1$
rv	Reverse String	$1 \rightarrow 1$
tb	Tabulate	$1 \rightarrow 1$
tl	TailN	$2 \rightarrow 1$
u	Unique	$1 \rightarrow 1$
un	UniqueN	$1 \rightarrow 1$
vH	WhichArr	$1 \rightarrow 1$
vW	SeqLen	$3 \rightarrow 1$
wh	Which	$1 \rightarrow 1$
zp	Zip	$2 \rightarrow 1$

7.5 Matrix

Symbol	Description	Stack Effect
Bi	IQR	$1 \rightarrow 1$
R9	Rot90	$1 \rightarrow 1$
Y!	Create Matrix	$3 \rightarrow 1$
Y*	Matrix Mult	$2 \rightarrow 1$
YM	Matrix ByRow	$3 \rightarrow 1$
mM	Model Matrix	$1 \rightarrow 1$
v2	Solve 2	$2 \rightarrow 1$
vL	Solve	$1 \rightarrow 1$
vQ	QR Decomp	$1 \rightarrow 1$
vV	SVD	$1 \rightarrow 1$
y!	Diag	$1 \rightarrow 1$
yD	Identity Matrix	$1 \rightarrow 1$
yc	EigenVectors	$1 \rightarrow 1$
yd	Determinant	$1 \rightarrow 1$
yf	Full	$1 \rightarrow 1$
yk	Kronecker	$2 \rightarrow 1$
yl	TriLower	$1 \rightarrow 1$

Symbol	Description	Stack Effect
ym	Create Matrix	3 $\rightarrow$ 1
yt	Trace	1 $\rightarrow$ 1
yu	TriUpper	1 $\rightarrow$ 1
yv	EigenValues	1 $\rightarrow$ 1

7.6 Statistics

Symbol	Description	Stack Effect
B5	FiveNum	1 $\rightarrow$ 1
BS	Summary	1 $\rightarrow$ 1
Ba	MAD	1 $\rightarrow$ 1
Bn	Min	1 $\rightarrow$ 1
Br	Range	1 $\rightarrow$ 1
Bx	Max	1 $\rightarrow$ 1
Ku	Kurtosis	1 $\rightarrow$ 1
Sd	CumSD	1 $\rightarrow$ 1
Sk	Skewness	1 $\rightarrow$ 1
Sm	MeanNA	1 $\rightarrow$ 1
Sn	SumNA	1 $\rightarrow$ 1
Sw	WeightedMean	2 $\rightarrow$ 1
V	Variance	1 $\rightarrow$ 1
h	Median	1 $\rightarrow$ 1
m	Mean	1 $\rightarrow$ 1
mn	Min2	2 $\rightarrow$ 1
mx	Max2	2 $\rightarrow$ 1
r1	Range1toN	1 $\rightarrow$ 1
s	Sum	1 $\rightarrow$ 1
sm	Summary	1 $\rightarrow$ 1
v	Variance	1 $\rightarrow$ 1
vB	PMin	2 $\rightarrow$ 1
vE	Row Sums	1 $\rightarrow$ 1
vG	PMax	2 $\rightarrow$ 1
vM	Col Means	1 $\rightarrow$ 1
vN	Row Means	1 $\rightarrow$ 1
vS	Col Sums	1 $\rightarrow$ 1

7.7 Statistical Modeling

Symbol	Description	Stack Effect
DN	Density Normal	$3 \rightarrow 1$
K!	Coef	$1 \rightarrow 1$
KB	BIC	$1 \rightarrow 1$
KC	Cor Test	$2 \rightarrow 1$
KE	Residuals	$1 \rightarrow 1$
KF	Fitted	$1 \rightarrow 1$
KL	LogLik	$1 \rightarrow 1$
KV	VCov	$1 \rightarrow 1$
av	Anova	$1 \rightarrow 1$
dB	Density Binomial	$3 \rightarrow 1$
dC	Density Chi-Square	$2 \rightarrow 1$
dG	Density Geometric	$2 \rightarrow 1$
dL	Density Logistic	$3 \rightarrow 1$
dN	Density Neg-Binomial	$3 \rightarrow 1$
dP	Density Poisson	$2 \rightarrow 1$
db	Density Beta	$3 \rightarrow 1$
dc	Density Cauchy	$3 \rightarrow 1$
de	Density Exponential	$2 \rightarrow 1$
df	Density F	$3 \rightarrow 1$
dg	Density Gamma	$3 \rightarrow 1$
dh	Density Hypergeometric	$4 \rightarrow 1$
dI	Density Log-Normal	$3 \rightarrow 1$
dn	Density Normal	$1 \rightarrow 1$
dt	Density Student-t	$2 \rightarrow 1$
du	Density Uniform	$3 \rightarrow 1$
dw	Density Weibull	$3 \rightarrow 1$
ka	Anova	$1 \rightarrow 1$
kb	Bartlett Test	$1 \rightarrow 1$
kc	Chi-Square Test	$1 \rightarrow 1$
ke	Predict	$2 \rightarrow 1$
kf	Fisher Test	$1 \rightarrow 1$
kg	GLM	$1 \rightarrow 1$
ki	AIC	$1 \rightarrow 1$
kk	KS Test	$2 \rightarrow 1$
kl	Linear Model	$1 \rightarrow 1$
kn	NLS	$1 \rightarrow 1$
ko	Loess	$1 \rightarrow 1$
kp	Prop Test	$2 \rightarrow 1$
kt	T-Test	$2 \rightarrow 1$
kv	F-Test Var	$2 \rightarrow 1$
kw	Wilcoxon Test	$2 \rightarrow 1$

Symbol	Description	Stack Effect
lm	Linear Model	1 $\rightarrow$ 1
mC	Conf Int	1 $\rightarrow$ 1
mL	LM Simple	2 $\rightarrow$ 1
mO	Offset	1 $\rightarrow$ 1
mP	Predict	1 $\rightarrow$ 1
mR	Model Frame	1 $\rightarrow$ 1
mT	Terms	1 $\rightarrow$ 1
mU	Update	2 $\rightarrow$ 1
mW	Formula	1 $\rightarrow$ 1
tt	T-Test	2 $\rightarrow$ 1
vd	Density	1 $\rightarrow$ 1
vp	PCA	1 $\rightarrow$ 1

7.8 Distributions & Tests

Symbol	Description	Stack Effect
BC	Covariance	2 $\rightarrow$ 1
Bc	Correlation	2 $\rightarrow$ 1
KK	Kruskal-Wallis	1 $\rightarrow$ 1
LG	Log Gamma	1 $\rightarrow$ 1
N	Random Normal	1 $\rightarrow$ 1
PN	Prob Normal	3 $\rightarrow$ 1
QN	Quantile Normal	3 $\rightarrow$ 1
RB	Random Beta	3 $\rightarrow$ 1
RE	Random Exponential	2 $\rightarrow$ 1
RN	Random Normal	3 $\rightarrow$ 1
RU	Random Uniform	3 $\rightarrow$ 1
cr	Correlation	2 $\rightarrow$ 1
cv	Covariance	2 $\rightarrow$ 1
di	Digamma	1 $\rightarrow$ 1
e-	Exp(x)-1	1 $\rightarrow$ 1
ex	Exp	1 $\rightarrow$ 1
ks	Shapiro-Wilk	1 $\rightarrow$ 1
l+	Log(1+x)	1 $\rightarrow$ 1
l1	Log 10	1 $\rightarrow$ 1
l2	Log 2	1 $\rightarrow$ 1
lb	Log Beta	2 $\rightarrow$ 1
lg	Log Natural	1 $\rightarrow$ 1
mB	Beta	2 $\rightarrow$ 1

Symbol	Description	Stack Effect
mG	Gamma	$1 \rightarrow 1$
pB	Prob Binomial	$3 \rightarrow 1$
pG	Prob Geometric	$2 \rightarrow 1$
pL	Prob Logistic	$3 \rightarrow 1$
pN	Prob Neg-Binomial	$3 \rightarrow 1$
pP	Prob Poisson	$2 \rightarrow 1$
pb	Prob Beta	$3 \rightarrow 1$
pc	Prob Cauchy	$3 \rightarrow 1$
pe	Prob Exponential	$2 \rightarrow 1$
pg	Prob Gamma	$3 \rightarrow 1$
ph	Prob Hypergeometric	$4 \rightarrow 1$
pl	Prob Log-Normal	$3 \rightarrow 1$
pn	Prob Normal	$1 \rightarrow 1$
ps	Psigamma	$2 \rightarrow 1$
pt	Prob Student-t	$2 \rightarrow 1$
pu	Prob Uniform	$3 \rightarrow 1$
pw	Prob Weibull	$3 \rightarrow 1$
qB	Quantile Binomial	$3 \rightarrow 1$
qG	Quantile Geometric	$2 \rightarrow 1$
qL	Quantile Logistic	$3 \rightarrow 1$
qN	Quantile Neg-Binomial	$3 \rightarrow 1$
qP	Quantile Poisson	$2 \rightarrow 1$
qb	Quantile Beta	$3 \rightarrow 1$
qc	Quantile Cauchy	$3 \rightarrow 1$
qe	Quantile Exponential	$2 \rightarrow 1$
qg	Quantile Gamma	$3 \rightarrow 1$
qh	Quantile Hypergeometric	$4 \rightarrow 1$
ql	Quantile Log-Normal	$3 \rightarrow 1$
qn	Quantile Normal	$1 \rightarrow 1$
qt	Quantile Student-t	$2 \rightarrow 1$
qu	Quantile Uniform	$3 \rightarrow 1$
qw	Quantile Weibull	$3 \rightarrow 1$
rB	Random Binomial	$3 \rightarrow 1$
rG	Random Geometric	$2 \rightarrow 1$
rL	Random Logistic	$3 \rightarrow 1$
rN	Random Neg-Binomial	$3 \rightarrow 1$
rP	Random Poisson	$2 \rightarrow 1$
rb	Random Binomial	$1 \rightarrow 1$
rc	Random Cauchy	$3 \rightarrow 1$
re	Random Exp	$1 \rightarrow 1$
rg	Random Gamma	$3 \rightarrow 1$

Symbol	Description	Stack Effect
rh	Random Hypergeometric	$4 \rightarrow 1$
r1	Random Log-Normal	$3 \rightarrow 1$
rp	Random Poisson	$1 \rightarrow 1$
rt	Random Student-t	$2 \rightarrow 1$
rw	Random Weibull	$3 \rightarrow 1$
s4	EmptyLog	$0 \rightarrow 1$
sR	Regexpr	$2 \rightarrow 1$
sX	Gregexpr	$2 \rightarrow 1$
tg	Trigamma	$1 \rightarrow 1$

7.9 Math Functions

Symbol	Description	Stack Effect
Bs	Standard Deviation	$1 \rightarrow 1$
SP	$\text{Sin}(\pi * x)$	$1 \rightarrow 1$
XC	CumSum	$1 \rightarrow 1$
a2	ArcTan2	$2 \rightarrow 1$
aC	ArcCos	$1 \rightarrow 1$
aS	ArcSin	$1 \rightarrow 1$
aT	ArcTan	$1 \rightarrow 1$
ac	ArcCosh	$1 \rightarrow 1$
as	ArcSinh	$1 \rightarrow 1$
at	ArcTanh	$1 \rightarrow 1$
bA	BitAnd	$2 \rightarrow 1$
c1	CumProd	$1 \rightarrow 1$
ch	Cosh	$1 \rightarrow 1$
cn	IsIn	$2 \rightarrow 1$
cp	$\text{Cos}(\pi * x)$	$1 \rightarrow 1$
mc	Cos	$1 \rightarrow 1$
ms	Sin	$1 \rightarrow 1$
mt	Tan	$1 \rightarrow 1$
sD	SetDiff	$2 \rightarrow 1$
sd	Standard Deviation	$1 \rightarrow 1$
sh	Sinh	$1 \rightarrow 1$
th	Tanh	$1 \rightarrow 1$
tp	$\text{Tan}(\pi * x)$	$1 \rightarrow 1$
vF	Diff	$1 \rightarrow 1$
vI	CumMin	$1 \rightarrow 1$
vX	CumMax	$1 \rightarrow 1$

7.10 Combinatorics & Special

Symbol	Description	Stack Effect
MC	Choose	$2 \rightarrow 1$
Xn	Choose (nCr)	$2 \rightarrow 1$
fp	Factorial	$1 \rightarrow 1$

7.11 Complex Numbers

Symbol	Description	Stack Effect
cA	Arg	$1 \rightarrow 1$
cI	Imag Part	$1 \rightarrow 1$
cJ	Conjugate	$1 \rightarrow 1$
cM	Modulus	$1 \rightarrow 1$
cR	Real Part	$1 \rightarrow 1$

7.12 String Operations

Symbol	Description	Stack Effect
,	Concat Vectors	$2 \rightarrow 1$
C	Concat	$2 \rightarrow 1$
J	Join List	$2 \rightarrow 1$
S	Split String	$2 \rightarrow 1$
S!	Translate	$3 \rightarrow 1$
SG	GSub	$3 \rightarrow 1$
Xc	Char Translate	$3 \rightarrow 1$
Xt	ToLower	$1 \rightarrow 1$
a	To String	$1 \rightarrow 1$
c	To Char	$1 \rightarrow 1$
dS	Split	$2 \rightarrow 1$
dU	Unsplit	$2 \rightarrow 1$
j	Join	$2 \rightarrow 1$
sB	Grepl	$2 \rightarrow 1$
sG	Grep	$2 \rightarrow 1$
sL	ToLower	$1 \rightarrow 1$
sM	Message	$1 \rightarrow 0$
sQ	EmptyChar	$0 \rightarrow 1$

Symbol	Description	Stack Effect
sS	Stop	1 $\rightarrow$ 0
sU	ToUpper	1 $\rightarrow$ 1
sW	Warning	1 $\rightarrow$ 0
sf	Sprintf	2 $\rightarrow$ 1
sn	NChar	1 $\rightarrow$ 1
ss	Substr	3 $\rightarrow$ 1
st	TrimWS	1 $\rightarrow$ 1

7.13 Set Operations

Symbol	Description	Stack Effect
sF	SetPrompt	1 $\rightarrow$ 0
sI	Intersect	2 $\rightarrow$ 1
sJ	SetWD	1 $\rightarrow$ 0
sN	Union	2 $\rightarrow$ 1
sY	SetScipen	1 $\rightarrow$ 0
se	Setenv	1 $\rightarrow$ 0

7.14 Bitwise Operations

Symbol	Description	Stack Effect
bL	BitShiftL	2 $\rightarrow$ 1
bN	BitNot	1 $\rightarrow$ 1
bO	BitOr	2 $\rightarrow$ 1
bR	BitShiftR	2 $\rightarrow$ 1
bX	BitXor	2 $\rightarrow$ 1

7.15 Type & Introspection

Symbol	Description	Stack Effect
AS	As Type	2 $\rightarrow$ 1
Xq	Is Prime	1 $\rightarrow$ 1
is	Is Type	2 $\rightarrow$ 1
mp	Is Prime	1 $\rightarrow$ 1
oA	Attr	2 $\rightarrow$ 1

Symbol	Description	Stack Effect
oC	Colnames	1 $\rightarrow$ 1
oa	Attributes	1 $\rightarrow$ 1
oc	Class	1 $\rightarrow$ 1
od	Dim	1 $\rightarrow$ 1
on	Names	1 $\rightarrow$ 1
or	Rownames	1 $\rightarrow$ 1
ot	Typeof	1 $\rightarrow$ 1
ou	Unlist	1 $\rightarrow$ 1

7.16 File I/O

Symbol	Description	Stack Effect
F	Read File	1 $\rightarrow$ 1
FL	List Files	1 $\rightarrow$ 1
FW	Write File	2 $\rightarrow$ 0
W	Write File	2 $\rightarrow$ 0
fA	Dirname	1 $\rightarrow$ 1
fC	Write CSV	2 $\rightarrow$ 0
fD	List Dirs	1 $\rightarrow$ 1
fN	File Create	1 $\rightarrow$ 1
fP	Abs Path	1 $\rightarrow$ 1
fT	Temp File	0 $\rightarrow$ 1
fX	Dir Exists	1 $\rightarrow$ 1
fb	Basename	1 $\rightarrow$ 1
fc	Read CSV	1 $\rightarrow$ 1
fd	Dir Create	1 $\rightarrow$ 1
fe	File Exists	1 $\rightarrow$ 1
fi	File Info	1 $\rightarrow$ 1
fm	File Remove	1 $\rightarrow$ 1
fr	Read Table	1 $\rightarrow$ 1
ft	Temp Dir	0 $\rightarrow$ 1
fw	Write Table	2 $\rightarrow$ 0

7.17 System

Symbol	Description	Stack Effect
T	Time Epoch	$0 \rightarrow 1$
ZT	Timezone	$0 \rightarrow 1$
ge	Getenv	$1 \rightarrow 1$
pi	GetPID	$0 \rightarrow 1$
sK	Toc	$1 \rightarrow 1$
sP	Sleep	$1 \rightarrow 0$
sT	Tic	$0 \rightarrow 1$
sV	RVersion	$0 \rightarrow 1$
um	Umask	$1 \rightarrow 1$
zg	GC	$0 \rightarrow 1$
zl	Locale	$0 \rightarrow 1$
zo	Options	$0 \rightarrow 1$
zt	SysTime	$0 \rightarrow 1$
zv	Version	$0 \rightarrow 1$

7.18 Other

Symbol	Description	Stack Effect
#	Filter	$2 \rightarrow 1$
&	Outer Product	$2 \rightarrow 1$
..	Pair	$2 \rightarrow 1$
A	To ASCII	$1 \rightarrow 1$
BT	Bingo Twin	$1 \rightarrow 1$
Bz	Scale	$1 \rightarrow 1$
Fa	Apply	$3 \rightarrow 1$
Ff	Filter (Func)	$2 \rightarrow 1$
F1	Lapply	$2 \rightarrow 1$
Fm	Mapply	$1 \rightarrow 1$
Fn	Find (Func)	$2 \rightarrow 1$
Fp	Position	$2 \rightarrow 1$
Fs	Sapply	$2 \rightarrow 1$
Fv	Vapply	$3 \rightarrow 1$
In	Inf	$0 \rightarrow 1$
KA	AOV	$1 \rightarrow 1$
Na	NA	$0 \rightarrow 1$
P	Product	$1 \rightarrow 1$
Pi	Pi	$0 \rightarrow 1$
Q	Quantile	$1 \rightarrow 1$
XP	Prime Factors	$1 \rightarrow 1$

Symbol	Description	Stack Effect
Xb	To Binary	1 $\rightarrow$ 1
Xd	To Date	1 $\rightarrow$ 1
Xk	RLE	1 $\rightarrow$ 2
Xo	Mode	1 $\rightarrow$ 1
Xr	RMS	1 $\rightarrow$ 1
Xs	Std Dev	1 $\rightarrow$ 1
dA	Aggregate	3 $\rightarrow$ 1
dE	Cut	2 $\rightarrow$ 1
dO	Order	1 $\rightarrow$ 1
dT	Table	1 $\rightarrow$ 1
dX	XTabs	2 $\rightarrow$ 1
dY	By	3 $\rightarrow$ 1
fa	Factors	1 $\rightarrow$ 1
fn	Negate	1 $\rightarrow$ 1
g	GCD	2 $\rightarrow$ 1
lA	Any	1 $\rightarrow$ 1
lX	Xor	2 $\rightarrow$ 1
lZ	NNZ	1 $\rightarrow$ 1
lc	LCM	2 $\rightarrow$ 1
mF	Factor	1 $\rightarrow$ 1
mI	IntDiv	2 $\rightarrow$ 1
mQ	Quantile	2 $\rightarrow$ 1
mf	Prime Factors	1 $\rightarrow$ 1
ml	LCM	2 $\rightarrow$ 1
n	To Number	1 $\rightarrow$ 1
ns	NegSlice	2 $\rightarrow$ 1
p	Print	1 $\rightarrow$ 0
pC	Prob Chi-Square	2 $\rightarrow$ 1
pf	Prob F	3 $\rightarrow$ 1
qC	Quantile Chi-Square	2 $\rightarrow$ 1
qf	Quantile F	3 $\rightarrow$ 1
r	Random (n)	1 $\rightarrow$ 1
r2	Rep	2 $\rightarrow$ 1
rC	Random Chi-Square	2 $\rightarrow$ 1
rf	Random F	3 $\rightarrow$ 1
rm	RMS	1 $\rightarrow$ 1
sC	Cat	1 $\rightarrow$ 0
sE	Date	0 $\rightarrow$ 1
sH	GetWD	0 $\rightarrow$ 1
sZ	EmptyNum	0 $\rightarrow$ 1
sc	Scale	1 $\rightarrow$ 1

Symbol	Description	Stack Effect
sr	Sub	$3 \rightarrow 1$
vA	All	$1 \rightarrow 1$
vC	Col Indices	$1 \rightarrow 1$
vD	Drop	$1 \rightarrow 1$
vR	Row Indices	$1 \rightarrow 1$
vT	RepMat	$3 \rightarrow 1$
zd	To Date	$1 \rightarrow 1$
\ 	Or	$2 \rightarrow 1$